PHYLOGENETIC TREE CONSTRUCTION USING MULTIPLE SEQUENCE ALIGNMENT (MSA)

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MINOR PROJECT

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ABSTRACT

Using phylogenetic tree, here I am trying to find the evolutionary connection and shared ancestry between different organisms (here I have used protein datasets of 7 different organisms). For this process to be done, first took the datasets of protein of 7 different organisms from NCBI protein and combined all into a single text document "cytochrome_sequences.fasta". These sequences are uploaded into MAFFT to get it aligned sequences are used to construct a phylogenetic tree based on the neighbour-joining method with 1500 bootstraps. The tree is visualised and annotated using iTOL, a web-based tool. The sequences is separated into two clades and all nodes showed bootstrap values 89%-100%, indicating that the tree topology is highly reliable. Drosophila melanogaster and Parachaetomium inaequale formed the most closely related pair, while Homo sapiens diverged earlier, when compared among the 7 different organisms. The result show that cytochrome c is highly conserved protein and is useful for studying evolutionary relationships among organisms and also shows how Multiple Sequence Alignment (MSA) and phylogenetic trees reveal evolutionary relationships.

INTRODUCTION

Evolution is the process by which all living things change over time through inherited genetic modifications and descent from common ancestors. We study evolution and the evolutionary relationships because it gives us a clear map of life on Earth and solves critical real-world problems.

According to this report we are focusing on phylogenetic trees. Basically, phylogenetic trees are a branching diagram that shows the evolutionary relationships and history among different biological species, groups, or genes based on similarities and differences in their physical or genetic traits. It acts as a hypothesis of how organisms descended from common ancestors. The starting point (base) of a rooted tree that represents the single common ancestor of all the organisms or taxa is the root of the tree. The points where a lineage splits into two or more daughter lines is the node (branch points) of the tree. Two lineages that branch from the same node and share an immediate common ancestor, making them each other's closest relatives represents the sister taxa. A grouping that includes a common ancestor and all of its descendants (living or extinct), forming a complete branch on the tree is the clade. Branches (Edges)- The lines that connect the nodes and tips. They represent evolutionary lineages. The endpoints at the very end of the branches are tips(leaves). They represent living or extinct species, populations, or taxonomic units.

Multiple Sequence Alignment (MSA) is the process of arranging three or more biological sequences (such as DNA, RNA, or protein) in a matrix to line up similar or evolutionarily related regions. It shows conserved regions (similar across species) and variable regions (different), which is the data the tree is built from. The tool used here is MAFFT (Multiple Alignment using Fast Fourier Transform).

There are two popular methods for building phylogenetic trees- Neighbour joining (Distance-based method) and Maximum likelihood (Character-based method). Neighbour joining method is preferred over Maximum likelihood because it is very fast and computationally inexpensive.

Cytochrome C is an essential protein in the electron transport chain, found in almost all organisms. It is highly conserved, so it is a classic marker for studying evolution.

The aim of this project is to retrieve homologous sequences, perform multiple sequence alignment, construct a Neighbour-Joining phylogenetic tree, and interpret the evolutionary relationships among the selected species.

METHODOLOGY

Sequence retrieval is done from NCBI protein, compiled all the datasets into a single document saved it as ".fasta" format. Here I have taken 7 datasets from different organisms. Then the sequence is aligned by using MAFFT. Phylogenetic tree is formed by using MEGA software and Neighbour-joining method as they are fast and distance-based. MEGA software provides bootstrap value to the tree. Then visualisation is done by iTOL. This provides some visualization effects to the tree.

SEQUENCES USED

No.	Species	Common Name	Protein Name	Accession ID	Length
1	Drosophila melanogaster	Fruit fly	cytochrome c	NP_001285984.1	108 aa
2	Arabidopsis thaliana	Plant	cytochrome c	AAB72175.1	114 aa
3	Trypanosoma grayi	Protist	cytochrome c	XP_009316014.1	114 aa
4	Homo sapiens	Human	cytochrome c	NP_061820.1	105 aa
5	Equus caballus	Horse	cytochrome c	NP_001157486.2	105 aa
6	Bos taurus	Cow	cytochrome c	NP_001039526.1	106 aa
7	Parachaetomium inaequale	Fungus	cytochrome c	KAK4038701.1	108 aa

RESULTS

Seven cytochrome c sequences were aligned using MAFFT. The three mammalian sequences (human, horse, cow) were nearly identical, while *Trypanosoma grayi* and *Arabidopsis thaliana* showed the most differences, including gaps near the N-terminus.

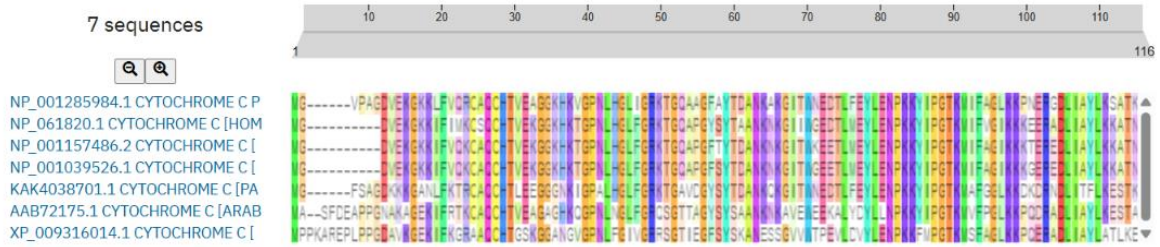


Figure1

The Neighbour-Joining tree separated the sequences into two main groups. The first group contained *Drosophila melanogaster* with *Parachaetomium inaequale* (bootstrap 100) and *Arabidopsis thaliana* with *Trypanosoma grayi* (bootstrap 89); these two pairs joined with bootstrap support of 100. The second group contained the three mammals: horse and cow formed a pair (bootstrap 92), with human as their nearest relative.

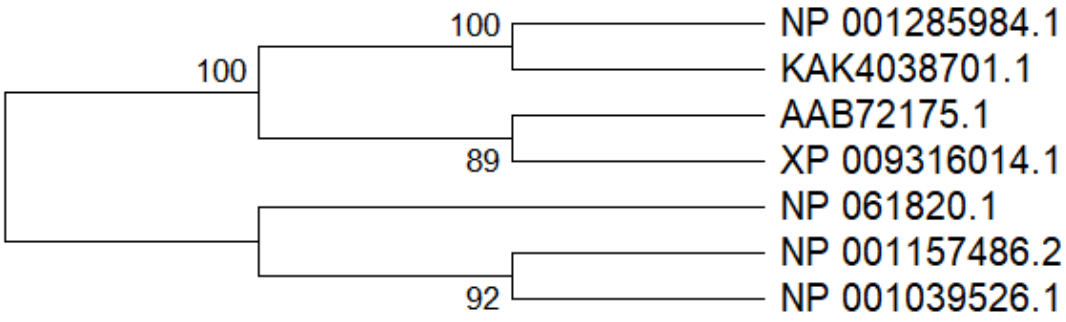
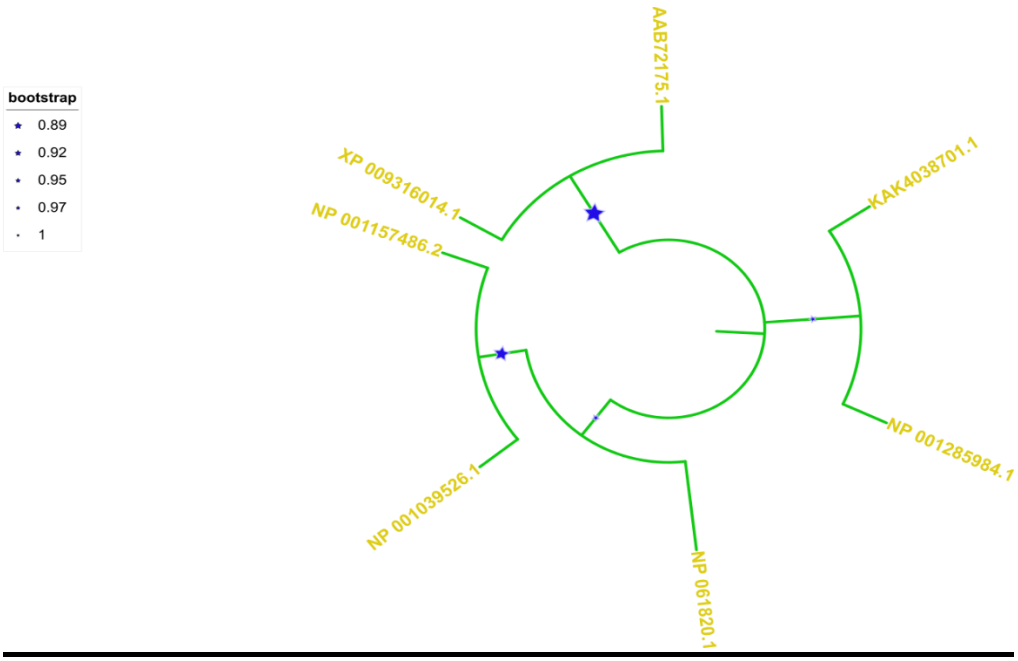


Figure2: Neighbour-Joining tree of cytochrome c from seven species. Numbers at the nodes are bootstrap values (1500 replicates)

[Done in MEGA – tree building]



[Done in iTOL – tree visualisation]

BOOTSTRAP SUPPORT

Node	Bootstrap (%)
Drosophila + Parachaetomium	100
Arabidopsis + Trypanosoma	89
Both pairs joined	100
Horse + Cow	92

All nodes had bootstrap support of 89% or higher.

CONCLUSION

In this project, cytochrome c protein sequences from seven species (*Drosophila melanogaster*, *Homo sapiens*, *Equus caballus*, *Bos taurus*, *Parachaetomium inaequale*, *Arabidopsis thaliana* and *Trypanosoma grayi*) were retrieved from NCBI, aligned using MAFFT, and used to build a Neighbor-Joining phylogenetic tree in MEGA with 1500 bootstrap replicates. The alignment showed large conserved regions, including the heme-binding motif, which reflects the essential role of cytochrome c. The tree separated the sequences into two well-supported groups (bootstrap 89-100%), and the three mammals (human, horse and cow) clustered together, matching their known evolutionary relationships. Some deeper groupings did not match the expected species relationships, which is likely because cytochrome c is short and highly conserved. Overall, the project shows how multiple sequence alignment and phylogenetic tree construction can be used to study evolutionary relationships, and that results are most reliable for closely related species. Using more genes, more species and Maximum Likelihood analysis would improve the tree.

TOOLS USED

1. NCBI	FASTA Sequence
2. MAFFT	Aligned sequences
3. MEGA	Phylogenetic tree
4. iTOL	Visualisation